PHYS-1310
EXAM 03
Given Time: 2 hrs

Name: _____ Student ID: _____

TABLE 18.1 Average Expansion Coefficients for Some Materials Near Room Temperature

Material (Solids)	Average Linear Expansion Coefficient $(\alpha)(°C)^{-1}$	Material (Liquids and Gases)	Average Volume Expansion Coefficient $(\beta)(°C)^{-1}$
Aluminum	24×10^{-6}	Acetone	1.5×10^{-4}
Brass and bronze	19×10^{-6}	Alcohol, ethyl	1.12×10^{-4}
Concrete	12×10^{-6}	Benzene	1.24×10^{-4}
Copper	17×10^{-6}	Gasoline	9.6×10^{-4}
Glass (ordinary)	9×10^{-6}	Glycerin	4.85×10^{-4}
Glass (Pyrex)	3.2×10^{-6}	Mercury	1.82×10^{-4}
Invar (Ni–Fe alloy)	0.9×10^{-6}	Turpentine	9.0×10^{-4}
Lead	29×10^{-6}	Air[a] at 0°C	3.67×10^{-3}
Steel	11×10^{-6}	Helium[a]	3.665×10^{-3}

[a]Gases do not have a specific value for the volume expansion coefficient because the amount of expansion depends on the type of process through which the gas is taken. The values given here assume the gas undergoes an expansion at constant pressure.

Hint Problem 20: Stefans law $P = \sigma A e T^4$ where P is the power in watts of electromagnetic waves radiated from the surface of the object, σ is a constant equal to 5.669 6 x 10^{28} W/m^2. K^4, A is the surface area of the object in square meters, e is the **emissivity,** and T is the surface temperature in kelvins.

For 28-31, root mean square velocity

$$v_{\text{rms}} = \sqrt{\overline{v^2}} = \sqrt{\frac{3k_B T}{m_0}} = \sqrt{\frac{3RT}{M}}$$

where M is the molar mass in kilograms per mole and is equal to $m_0 N_A$

$$k_B = \frac{R}{N_A} \qquad 1.380\,648\,52\ (79) \times 10^{-23}\ \text{J/K}$$

$$R \qquad 8.314\,459\,8\ (48)\ \text{J/mol}\cdot\text{K}$$

Coefficient of performance

$$\text{COP (cooling mode)} = \frac{\text{energy transferred at low temperature}}{\text{work done on heat pump}} = \frac{|Q_c|}{W}$$

TABLE 19.2 Latent Heats of Fusion and Vaporization

Substance	Melting Point (°C)	Latent Heat of Fusion (J/kg)	Boiling Point (°C)	Latent Heat of Vaporization (J/kg)
Helium[a]	−272.2	5.23×10^3	−268.93	2.09×10^4
Oxygen	−218.79	1.38×10^4	−182.97	2.13×10^5
Nitrogen	−209.97	2.55×10^4	−195.81	2.01×10^5
Ethyl alcohol	−114	1.04×10^5	78	8.54×10^5
Water	0.00	3.33×10^5	100.00	2.26×10^6
Sulfur	119	3.81×10^4	444.60	3.26×10^5
Lead	327.3	2.45×10^4	1 750	8.70×10^5
Aluminum	660	3.97×10^5	2 450	1.14×10^7
Silver	960.80	8.82×10^4	2 193	2.33×10^6
Gold	1 063.00	6.44×10^4	2 660	1.58×10^6
Copper	1 083	1.34×10^5	1 187	5.06×10^6

[a]Helium does not solidify at atmospheric pressure. The melting point given here corresponds to a pressure of 2.5 MPa.

TABLE 19.1 Specific Heats of Some Substances at 25°C and Atmospheric Pressure

Substance	Specific Heat (J/kg · °C)	Substance	Specific Heat (J/kg · °C)
Elemental solids		*Other solids*	
Aluminum	900	Brass	380
Beryllium	1 830	Glass	837
Cadmium	230	Ice (−5°C)	2 090
Copper	387	Marble	860
Germanium	322	Wood	1 700
Gold	129	*Liquids*	
Iron	448	Alcohol (ethyl)	2 400
Lead	128	Mercury	140
Silicon	703	Water (15°C)	4 186
Silver	234	*Gas*	
		Steam (100°C)	2 010

Note: To convert values to units of cal/g · °C, divide by 4 186.

General Doppler Shift Expression

$$f' = \left(\frac{v + v_O}{v - v_S}\right) f$$

Sound level

$$\beta \equiv 10 \log \left(\frac{I}{I_0}\right)$$

1. *A guitar string fixed between two points is under a tension of 12.0 N. A frequency of 70.0 Hz excites the fundamental. If the tension were increased to 30.0 N, what would be the new fundamental frequency?
   A. Additional information is needed. B. 111 Hz C. 175 Hz D. 28 Hz E. 44 Hz
2. *A student sounds a 300 Hz tuning fork and spins it overhead in a circle of radius 0.50 m and with a period of 0.70 s. What frequency do others hear at instants when the tuning fork is directly approaching them?
   A. 300 Hz B. 296 Hz C. 308 Hz D. 292 Hz E. 304 Hz

3. Markings to indicate length are placed on a steel tape in a room that is at a temperature of 22°C. Measurements are then made with the same tape on a day when the temperature is 27°C. Assume the objects you are measuring have a smaller coefficient of linear expansion than steel. Are the measurements
   A. too long, B. too short, or C. accurate?
4. When a certain gas under a pressure of 5.00×10^6 Pa at 25.0°C is allowed to expand to 3.00 times its original volume, its final pressure is 1.07×10^6 Pa. What is its final temperature?
   A. 450 K B. 233 K C. 212 K D. 191 K E. 115 K
5. If the volume of an ideal gas is doubled while its temperature is quadrupled, does the pressure
   A. remain the same, B. decrease by a factor of 2, C. decreases by a factor of 4,
   D. increase by a factor of 2, or E. increases by a factor of 4

6. The pendulum of a certain pendulum clock is made of brass. When the temperature increases, what happens to the period of the clock?
   A. It increases. B. It decreases. C. It remains the same.

7. A temperature of 162°F is equivalent to what temperature in Kelvins?
   A. 373 K B. 288 K C. 345 K D. 201 K E. 308 K
8. A cylinder with a piston holds 0.50 m^3 of oxygen at an absolute pressure of 4.0 atm. The piston is pulled outward, increasing the volume of the gas until the pressure drops to 1.0 atm. If the temperature stays constant, what new volume does the gas occupy?
   A. 1.0 m^3 B. 1.5 m^3 C. 2.0 m^3 D. 0.12 m^3 E. 2.5 m^3

9. What would happen if the glass of a thermometer expanded more on warming than did the liquid in the tube?
   A. The thermometer would break.

B. It could be used only for temperatures below room temperature.
C. You would have to hold it with the bulb on top.
D. The scale on the thermometer is reversed so that higher temperature values would be found closer to the bulb.
E. The numbers would not be evenly spaced.

10. Two cylinders A and B at the same temperature contain the same quantity of the same kind of gas. Cylinder A has three times the volume of cylinder B. What can you conclude about the pressures the gases exert?
A. We can conclude nothing about the pressures.
B. The pressure in A is three times the pressure in B.
C. The pressures must be equal.
D. The pressure in A must be one-third the pressure in B.

11. Suppose you empty a tray of ice cubes into a bowl partly full of water and cover the bowl. After one-half hour, the contents of the bowl come to thermal equilibrium, with more liquid water and less ice than you started with. Which of the following is true?
A. The temperature of the liquid water is higher than the temperature of the remaining ice.
B. The temperature of the liquid water is the same as that of the ice.
C. The temperature of the liquid water is less than that of the ice.
D. The comparative temperatures of the liquid water and ice depend on the amounts present.

12. A hole is drilled in a metal plate. When the metal is raised to a higher temperature, what happens to the diameter of the hole?
A. It decreases.
B. It increases.
C. It remains the same.
D. The answer depends on the initial temperature of the metal.
E. None of those answers is correct.

13. On a very cold day in upstate New York, the temperature is -25°C, which is equivalent to what Fahrenheit temperature?
A. -46°F B. -77°F C. 18°F D. -25°F E-13°F

14. Assume you are measuring the specific heat of a sample of originally hot metal by using a calorimeter containing water. Because your calorimeter is not perfectly insulating, energy can transfer by heat between the contents of the calorimeter and the room. To obtain the most accurate result for the specific heat of the metal, you should use water with which initial temperature?
A. slightly lower than room temperature
B. the same as room temperature
C. slightly higher than room temperature
D. whatever you like because the initial temperature makes no difference

15. An amount of energy is added to ice, raising its temperature from -10°C to 25°C. A larger amount of energy is added to the same mass of water, raising its temperature from 15°C to 20°C. From these results, what would you conclude?
A. Overcoming the latent heat of fusion of ice requires an input of energy.
B. The latent heat of fusion of ice delivers some energy to the system.
C. The specific heat of ice is less than that of water.
D. The specific heat of ice is greater than that of water.
E. More information is needed to draw any conclusion.

16. *How much energy is required to raise the temperature of 5.00 kg of lead from 20.0°C to its melting point of 327°C. The specific heat of lead is 128 J/kg °C.
A. 4.04×10^5 J B. 1.07×10^5 J C. 8.15×10^4 J D. 2.13×10^4 J
E. 1.96×10^5 J

17. Ethyl alcohol has about one-half the specific heat of water. Assume equal amounts of energy are transferred by heat into equal-mass liquid samples of alcohol and water in separate insulated containers. The water rises in temperature by 25°C. How much will the alcohol rise in temperature?
A. It will rise by 12°C.
B. It will rise by 25°C.
C. It will rise by 50°C.
D. It depends on the rate of energy transfer.
E. It will not rise in temperature.

18. The specific heat of substance A is greater than that of substance B. Both A and B are at the same initial temperature when equal amounts of energy are added to them. Assuming no melting or vaporization occurs, which of the following can be concluded about the final temperature T_A of substance A and the final temperature T_B of substance B?
A. $T_A > T_B$
B. $T_A < T_B$
C. $T_A = T_B$
D. More information is needed.

19. *A 100-g piece of copper, initially at 95.0°C, is dropped into 200 g of water contained in a 280-g aluminum can; the water and can are initially at 15.0°C. What is the final temperature of the system? (Specific heats of copper and aluminum are 0.092 and 0.215 cal/g °C, respectively.)
A. 16°C
B. 18°C
C. 24°C
D. 26°C
E. none of those answers

20. Star *A* has twice the radius and twice the absolute surface temperature of star *B*. The emissivity of both stars can be assumed to be 1. What is the ratio of the power output of star *A* to that of star *B*?
A. 4 B. 8 C. 16 D. 32 E. 64

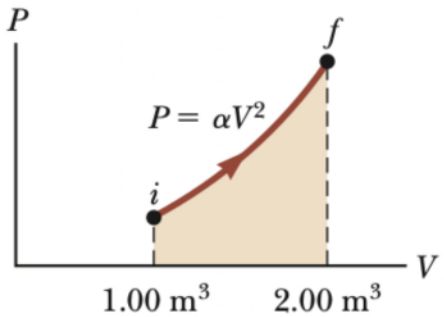
21. If a gas is compressed isothermally, which of the following statements is true?
A. Energy is transferred into the gas by heat.
B. No work is done on the gas.
C. The temperature of the gas increases.
D. The internal energy of the gas remains constant.
E. None of those statements is true.

22. When a gas undergoes an adiabatic expansion, which of the following statements is true?
A. The temperature of the gas does not change.
B. No work is done by the gas.
C. No energy is transferred to the gas by heat.
D. The internal energy of the gas does not change.
E. The pressure increases.

23. If a gas undergoes an isobaric process, which of the following statements is true?
   A. The temperature of the gas doesn't change.
   B. Work is done on or by the gas.
   C. No energy is transferred by heat to or from the gas.
   D. The volume of the gas remains the same.
   E. The pressure of the gas decreases uniformly.

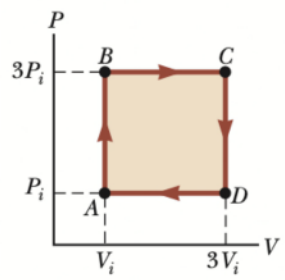
24. *How long would it take a 1 000 W heater to melt 1.00 kg of ice at 0.0°C, assuming all the energy from the heater is absorbed by the ice?
   A. 4.18 s B. 41.8 s C. 5.55 min D. 6.25 min E. 38.4 min

25. *An ideal gas is taken through a quasi-static process described by $P = aV^2$, with a = 5.00 atm/m^6, as shown in Figure. The gas is expanded to twice its original volume of 1.00 m^3. How much work is done on the expanding gas in this process?
   A. -1.18MJ
   B. 11.8MJ
   C. 118MJ
   D. 0.118MJ



26. *An ideal gas initially at P_i, V_i, and T_i is taken through a cycle as shown in Figure. What is the net energy added by heat to the gas per cycle?
   A. 9.08 KJ B. 908KJ C. 200KJ D. 90KJ

27. *An ideal gas initially at P_i , V_i , and T_i is taken through a cycle as shown in Figure. What is the net energy added by heat to the system per cycle?
   A. 4PiVi
   B. 6PiVi
   C. 2PiVi
   D. 5PiVi



28. Cylinder A contains oxygen (O_2) gas, and cylinder B contains nitrogen (N_2) gas. If the molecules in the two cylinders have the same rms speeds, which of the following statements is *false*?
   A. The two gases have different temperatures.
   B. The temperature of cylinder B is less than the temperature of cylinder A.
   C. The temperature of cylinder B is greater than the temperature of cylinder A.
   D. The average kinetic energy of the nitrogen molecules is less than the average kinetic energy of the oxygen molecules.

29. An ideal gas is maintained at constant pressure. If the temperature of the gas is increased from 200 K to 600 K, what happens to the rms speed of the molecules?
    A. It increases by a factor of 3.
    B. It remains the same.
    C. It is one-third the original speed.
    D. It is $\sqrt{3}$ times the original speed.
    E. It increases by a factor of 6.

30. Two samples of the same ideal gas have the same pressure and density. Sample B has twice the volume of sample A. What is the rms speed of the molecules in sample B?
    A. twice that in sample A
    B. equal to that in sample A
    C. half that in sample A
    D. impossible to determine

31. A gas is at 200 K. If we wish to double the rms speed of the molecules of the gas, to what value must we raise its temperature?
    A. 283 K B. 400 K C. 566 K D. 800 K E. 1130 K

32. Which of the assumptions below is *not* made in the kinetic theory of gases?
    A. The number of molecules is very large.
    B. The molecules obey Newton's laws of motion.
    C. The forces between molecules are long range.
    D. The gas is a pure substance.
    E. The average separation between molecules is large compared to their dimensions.

33. The second law of thermodynamics implies that the coefficient of performance of a refrigerator must be what?
    A. less than 1
    B. less than or equal to 1
    C. greater than or equal to 1
    D. finite
    E. greater than 0

34. Assume a sample of an ideal gas is at room temperature. What action will *necessarily* make the entropy of the sample increase?
    A. Transfer energy into it by heat.
    B. Transfer energy into it irreversibly by heat.
    C. Do work on it.
    D. Increase either its temperature or its volume, without letting the other variable decrease.
    E. None of those choices is correct.

35. A refrigerator has 18.0 kJ of work done on it while 115 kJ of energy is transferred from inside its interior. What is its coefficient of performance?
    A. 3.40 B. 2.80 C. 8.90 D. 6.40 E. 5.20

36. Of the following, which is *not* a statement of the second law of thermodynamics?
    A. No heat engine operating in a cycle can absorb energy from a reservoir and use it entirely to do work.

B. No real engine operating between two energy reservoirs can be more efficient than a Carnot engine operating between the same two reservoirs.
C. When a system undergoes a change in state, the change in the internal energy of the system is the sum of the energy transferred to the system by heat and the work done on the system.
D. The entropy of the Universe increases in all natural processes.
E. Energy will not spontaneously transfer by heat from a cold object to a hot object.

37. A compact air-conditioning unit is placed on a table inside a well-insulated apartment and is plugged in and turned on. What happens to the average temperature of the apartment?
A. It increases.
B. It decreases.
C. It remains constant.
D. It increases until the unit warms up and then decreases.
E. The answer depends on the initial temperature of the apartment.

38. *A steam turbine operates at a boiler temperature of 450 K and an exhaust temperature of 300 K. What is the maximum theoretical efficiency of this system?
A. 0.240 B. 0.500 C. 0.333 D. 0.667 E. 0.150

39. A thermodynamic process occurs in which the entropy of a system changes by 28 J/K. According to the second law of thermodynamics, what can you conclude about the entropy change of the environment?
A. It must be 18 J/K or less.
B. It must be between 18 J/K and 0.
C. It must be equal to 18 J/K.
D. It must be 18 J/K or more.
E. It must be zero.

40. An engine does 15.0 kJ of work while exhausting 37.0 kJ to a cold reservoir. What is the efficiency of the engine?
A. 0.150 B. 0.288 C. 0.333 D. 0.450 E. 1.20

41. Which statement is **false**?
A. The density of a gas is constant as long as its temperature remains constant.
B. Gases can be expanded without limit.
C. Gases diffuse into each other and mix almost immediately when put into the same container.
D. The molecular weight of a gaseous compound is a non-variable quantity.
E. Pressure must be exerted on a sample of a gas in order to confine it.

42. *A sample of oxygen occupies 47.2 liters under a pressure of 1240 torr at 25°C. What volume would it occupy at 25°C if the pressure were decreased to 730 torr?
A. 27.8 L B. 29.3 L C. 32.3 L D. 47.8 L E. 80.2 L

43. A sample of nitrogen occupies 5.50 liters under a pressure of 900 torr at 25°C. At what temperature will it occupy 10.0 liters at the same pressure?
A. 32°C B. -109°C C. 154°C D. 269°C E. 370°C

44. Under conditions of fixed temperature and amount of gas, Boyle's law requires that
I. $P_1V_1 = P_2V_2$
II. $PV = \text{constant}$
III. $P_1/P_2 = V_2/V_1$

A. I only
B. II only
C. III only
D. I, II, and III
E. another combination

45. The volume of a sample of nitrogen is 6.00 liters at 35°C and 740 torr. What volume will it occupy at STP?
A. 6.59 L B. 5.46 L C. 6.95 L D. 5.67 L E. 5.18 L

46. Which one of the following statements is **not consistent** with the kinetic-molecular theory of gases?
A. Individual gas molecules are relatively far apart.
B. The actual volume of the gas molecules themselves is very small compared to the volume occupied by the gas at ordinary temperatures and pressures.
C. The average kinetic energies of different gases are different at the same temperature.
D. There is no net gain or loss of the total kinetic (translational) energy in collisions between gas molecules.
E. The theory explains most of the observed behavior of gases at ordinary temperatures and pressures.

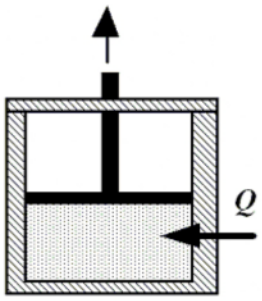
47. A real gas most closely approaches the behavior of an ideal gas under conditions of:
A. high P and low T
B. low P and high T
C. low P and T
D. high P and T
E. STP

48. The ideal gas law predicts that the molar volume (volume of one mole) of gas equals:
A. gRT/PV
B. (MW)P/RT
C. $1/2ms^{-2}$
D. RT/P
E. 22.4 L at any temperature and pressure

49. Three 1.0 liter flasks are filled with H_2, O_2 and Ne, respectively, at STP.
Which of the following statements is **true**?
A. Each flask has the same number of gas molecules.
B. The velocity of the gas molecules is the same in each flask.
C. The density of each gas is the same.
D. There are twice as many O_2 and H_2 molecules as Ne atoms.
E. None of the above is true.

50. For a gas, which pair of variables are inversely proportional to each other (if all other conditions remain constant)?
A. P, T B. P, V C. V, T D. n, V E. n, P

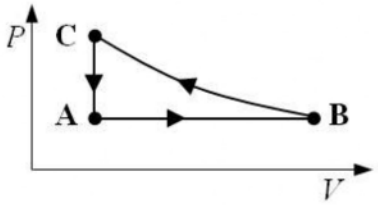
51. *When the gas enclosed beneath the piston shown in the figure receives 3000 J of heat, Q, from its surroundings, it performs 3840 J of work in raising the piston. What is the change in the internal energy of the gas?



A. +670 J
B. –670 J
C. –840 J
D. +5010 J
E. zero joules

Problems 52 and 53

An ideal monatomic gas expands isobarically from state **A** to state **B**. It is then compressed isothermally from state **B** to state **C** and finally cooled at constant volume until it returns to its initial state **A**.



$V_A = 4 \times 10^{-3}$ m^3 $\quad V_B = 8 \times 10^{-3}$ m^3
$P_A = P_B = 1 \times 10^6$ Pa $\quad T_A = 600$ K
$P_C = 2 \times 10^6$ Pa

52. *What is the temperature of the gas when it is in state **B**?
A. 437 K B. 573 K C. 927 K D. 1200 K E. 1473 K

53. *How much work is done by the gas in expanding isobarically from **A** to **B**?
A. 1×10^3 J B. 2×10^3 J C. 3×10^3 J D. 4×10^3 J E. 5×10^3 J

54. The decibel level of a jackhammer is 125 dB relative to the threshold of hearing. Determine the sound intensity produced by the jackhammer.
A. 1.0 W/m^2
B. 3.2 W/m^2
C. 4.8 W/m^2
D. 12 W/m^2
E. 88 W/m^2

55. *A train moving at a constant speed is passing a stationary observer on a platform. On one of the train cars, a flute player is continually playing the note known as concert A (f = 440 Hz). After the flute has passed, the observer hears the sound with a frequency of 415 Hz. What is the speed of the train? The speed of sound in air is 343 m/s.
A. 7.3 m/s B. 12 m/s C. 21 m/s D. 37 m/s E. 42 m/s

56. A wave has a frequency of 58 Hz and a speed of 31 m/s. What is the wavelength of this wave?
A. 1.9 m B. 3.5 m C. 0.29 m D. 0.53 m E. 31 m

57. Which one of the following waves is purely *longitudinal*?
A. sound waves in air

B. microwaves traveling through air
C. radio waves traveling through vacuum
D. waves on a plucked violin string
E. surface waves in a shallow pan of water

58. A periodic wave is produced on a stretched string. Which one of the following properties is *not* related to the speed of the wave?
A. frequency
B. amplitude
C. period
D. wavelength
E. tension in the string

59. What is the wavelength of a wave with a speed of 9.0 m/s and a period of 0.33 s?
A. 0.33 m B. 1.5 m C. 3.0 m D. 13 mE. 27 m

60. Which one of the following statements concerning waves is false?
A. A wave can have both transverse and longitudinal components.
B. A wave carries energy from one place to another.
C. A wave does not result in the bulk flow of the material of its medium.
D. A wave is a traveling disturbance.
E. A transverse wave is one in which the disturbance is parallel to the direction of travel.

61. The tension in a taut rope is increased by a factor of 9. How does the speed of wave pulses on the rope change, if at all?
A. The speed remains the same.
B. The speed is reduced by a factor of 3.
C. The speed is reduced by a factor of 9.
D. The speed is increased by a factor of 3.
E. The speed is increased by a factor of 9.

62. *A wave has an amplitude of 0.35 m, a frequency of 1.05×10^6 Hz, and travels in the positive x direction at the speed of light, 3.00×10^8 m/s. Which one of the following equations correctly represents this wave?
A. $y = 0.35 \sin (6.60 \times 10^6 t - 0.022x)$
B. $y = 0.35 \sin (6.60 \times 10^6 t + 0.022x)$
C. $y = 0.35 \sin (286t - 1.05 \times 10^6 x)$
D. $y = 0.35 \sin (286t + 1.05 \times 10^6 x)$
E. $y = 0.35 \sin (1.05 \times 10^6 t + 3.00 \times 10^8 x)$

63. *A transverse wave is traveling in the $-x$ direction on a string that has a linear density of 0.011 kg/m. The tension in the string is 7.4 N. The amplitude of the wave is 0.017 m; and its wavelength is 1.5 m. Which one of the following is the correct equation for the displacement of a particle from its equilibrium position?
A. $y = (0.017 \text{ m}) \sin [(84 \text{ Hz})t + (4.2 \text{ m}^{-1})x]$
B. $y = (0.017 \text{ m}) \sin [(110 \text{ Hz})t + (4.2 \text{ m}^{-1})x]$
C. $y = (4.2 \text{ m}) \sin [(84 \text{ Hz})t + (59 \text{ m}^{-1})x]$
D. $y = (4.2 \text{ m}) \sin [(110 \text{ Hz})t - (59 \text{ m}^{-1})x]$
E. $y = (0.017 \text{ m}) \sin [(110 \text{ Hz})t - (8.4 \text{ m}^{-1})x]$

64. *A transverse periodic wave on a string with a linear density of 0.200 kg/m is described by the following equation: $y = 0.08 \sin (469t - 28.0x)$, where x and y are in meters and t is in seconds. What is the tension in the string?
A. 3.99 N B. 32.5 N C. 56.1 N D. 65.8 N E. 79.6 N

65. Of the three traveling waves listed below, which one(s) is(are) traveling *in the +x direction*?

(1) $y = +3.2 \sin [4.1t + 2.3x]$
(2) $y =$ ñ$6.8 \sin [$ñ$3.0t + 1.5x]$
(3) $y = +4.9 \sin [12.0t + 18x]$

A. *1* only
B. *2* only
C. *3* only
D. *1*, *2,* and *3*
E. *2* and *3* only

66. *A transverse periodic wave is established on a string. The wave is described by the expression

$$y = 0.005 \sin (20.0x + 2\pi ft)$$

where y is in meters when x and t are in meters and seconds, respectively. If the wave travels with a speed of 20.0 m/s, what is its frequency, f?
A. 0.16 Hz B. 0.64 Hz C. 31.9 Hz D. 63.7 Hz E. 400 Hz

67. A transverse periodic wave on a string with a linear density of 0.200 kg/m is described by the following equation: $y = 0.08 \sin (469t$ ñ $28.0x)$, where x and y are in meters and t is in seconds. What is the tension in the string?
A. 3.99 N B. 32.5 N C. 56.1 N D. 65.8 N E. 79.6 N

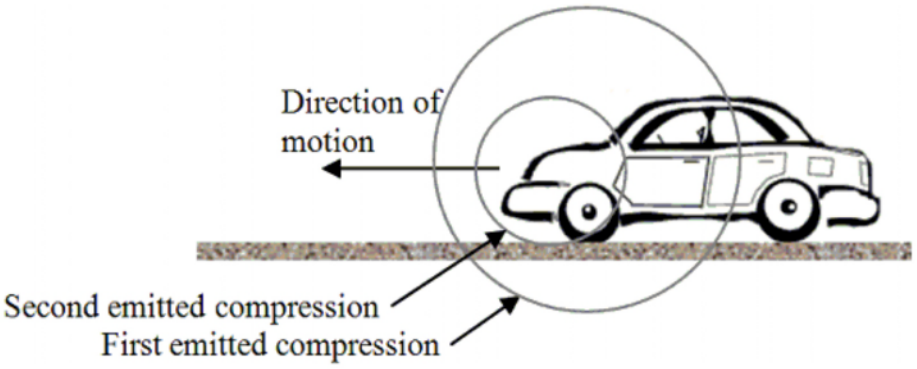
68. *A stationary railroad whistle is sounded. An echo is heard 5.0 seconds later by the train's engineer. If the speed of sound is 343 m/s, how far away is the reflecting surface?
A. 68 m B. 140 m C. 860 m D. 1700 m E. 2000 m

69. *Fans are watching a baseball game from different positions. One fan is located directly behind home plate, 18.3 m from the batter. The other fan is in the centerfield bleachers, 127 m from the batter. Both fans observe the batter strike the ball at the same time (because the speed of light is about a million times faster than that of sound), but the fan behind home plate hears the sound first. What is the time difference between hearing the sound at the two locations? Use 345 m/s as the speed of sound.
A. 0.316 s B. 0.368 s C. 3.17 s D. 1.89 s E. 0.053 s

70. *Two children setup a telephone by placing a long, slender aluminum ($Y = 6.9 \times 10^{10}$ N/m^2) rod that has a length of 6.1-m between their two houses. To communicate, a child taps a coded message on one end. How long do the sound waves take to reach the other end? Note: the density of aluminum is 2700 kg/m^3.
A. 1.2 ◊ 10^{ñ3} s
B. 2.4 ◊ 10^{ñ7} s
C. 8.0 ◊ 10^{ñ6} s
D. 6.3 ◊ 10^{ñ5} s
E. 1.1 ◊ 10^{ñ4} s

71. A bell emits sound energy uniformly in all directions at a rate of 4.00 x $10^{ñ3}$ W. What is the intensity of the wave 100.0 m from the bell?
   A. 3.18 x $10^{ñ8}$ W/m^2
   B. 3.14 x $10^{ñ7}$ W/m^2
   C. 5.02 x $10^{ñ2}$ W/m^2
   D. 5.02 x 10^2 W/m^2
   E. 6.28 x 10^7 W/m^2

The car in the drawing is moving to the left at 35 m/s. The car's horn continuously emits a 2.20 ◊ 10^2 Hz sound. The figure also shows the first two regions of compression of the emitted sound waves. The speed of sound is 343 m/s.



72. *How far does the car move in one period of the sound emitted from the horn?
   A. 0.08 m B. 0.16 m C. 8 m D. 16 m E. 35 m

73. *What change in entropy occurs when a 27.9-g ice cube at 212°C is transformed into steam at 115°C? Need

Specific Heat (J/kg ? °C) : Ice (25°C)- 2090; Water (15°C) -4186
Latent Heat of Vaporization (J/kg) 2.26 x 10^6
Latent Heat of Fusion (J/kg) 3.33 x 10^5

   A. 244J/K B. 100J/K C. 24J/k D. 2440J/K

74. *Calculate the change in entropy of 250 g of water warmed slowly from 20.0°C to 80.0°C
   A. 195 J/K B. 1.95J/K C. 19.5J/K D. 1950J/K

75. *A sample of a diatomic ideal gas has pressure *P* and volume *V*. When the gas is warmed, its pressure triples and its volume doubles. This warming process includes two steps, the first at constant pressure and the second at constant volume. Determine the amount of energy transferred to the gas by heat.
   A. 13.5PV B. 1.35PV C. 135PV D. 1350PV

Name: _____ Student ID: _____

Multiple Choice

* 1.	_____	11.	_____
* 2.	_____	12.	_____
3.	_____	13.	_____
4.	_____	14.	_____
5.	_____	15.	_____
6.	_____	16.	_____
7.	_____	17.	_____
8.	_____	18.	_____
9.	_____	* 19.	_____
10.	_____	20.	_____

21.	_____	31.	_____
22.	_____	32.	_____
23.	_____	33.	_____
* 24.	_____	34.	_____
* 25.	_____	35.	_____
* 26.	_____	36.	_____
* 27.	_____	37.	_____
28.	_____	* 38.	_____
29.	_____		
30.	_____		

NOTE: Questions with * have 2 points each, the rests hold 1 point.

39. _____
40. _____
41. _____
* 42. _____
43. _____
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67. _____
* 68. _____
* 69. _____
* 70. _____
* 71. _____
* 72. _____
* 73. _____
* 74. _____
* 75. _____

NOTE: Questions with * have 2 points each, the rests hold 1 point.